An Udderly Exciting Analysis: Exploring bovine milk microbiota throughout skim milk powder processing

Kailyn Hanke

BIOMI 6300

I analyzed data from a previously published paper!

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RESEARCH ARTICLE
Applied and Environmental Science

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Tracking the Dairy Microbiota from Farm Bulk Tank to Skimmed Milk Powder

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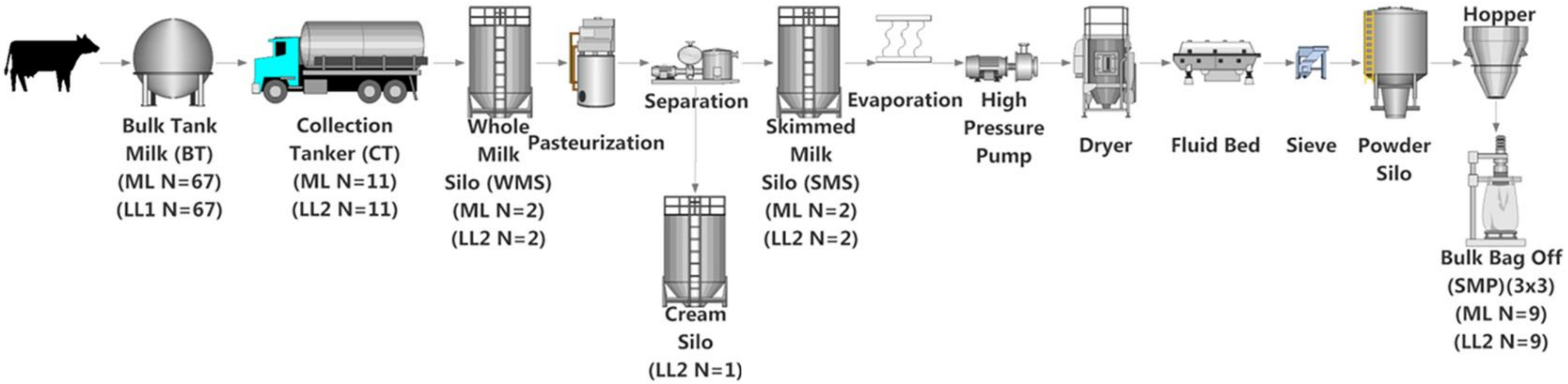
- Used 16S rRNA gene sequencing
- Tracked the microbiota of raw milks on farms to a final skimmed milk powder product

Six different types of dairy samples were collected



(McHugh et al., 2020)

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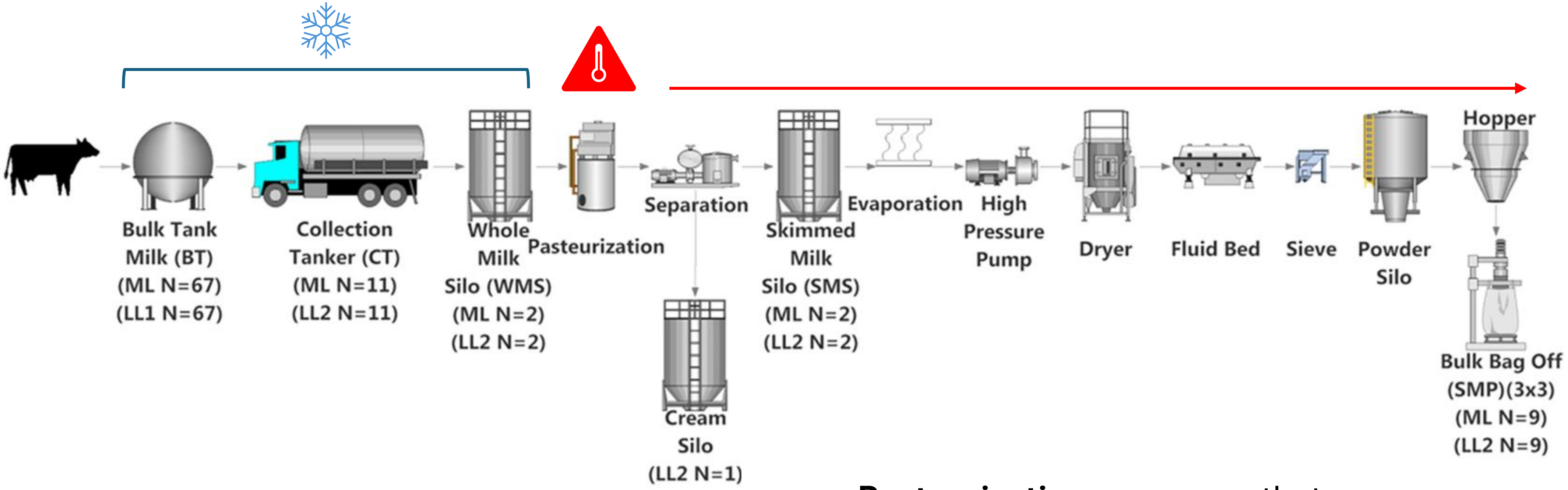


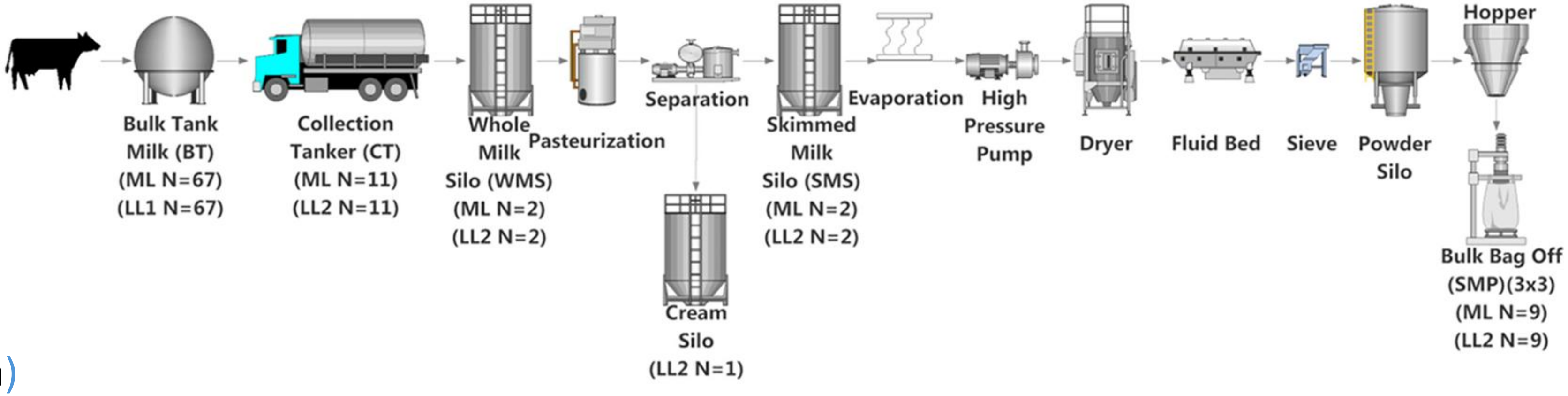
Pasteurization: a process that applies heat to destroy pathogens in foods

(McHugh et al., 2020)

Samples were collected during two "seasons"

Seasons

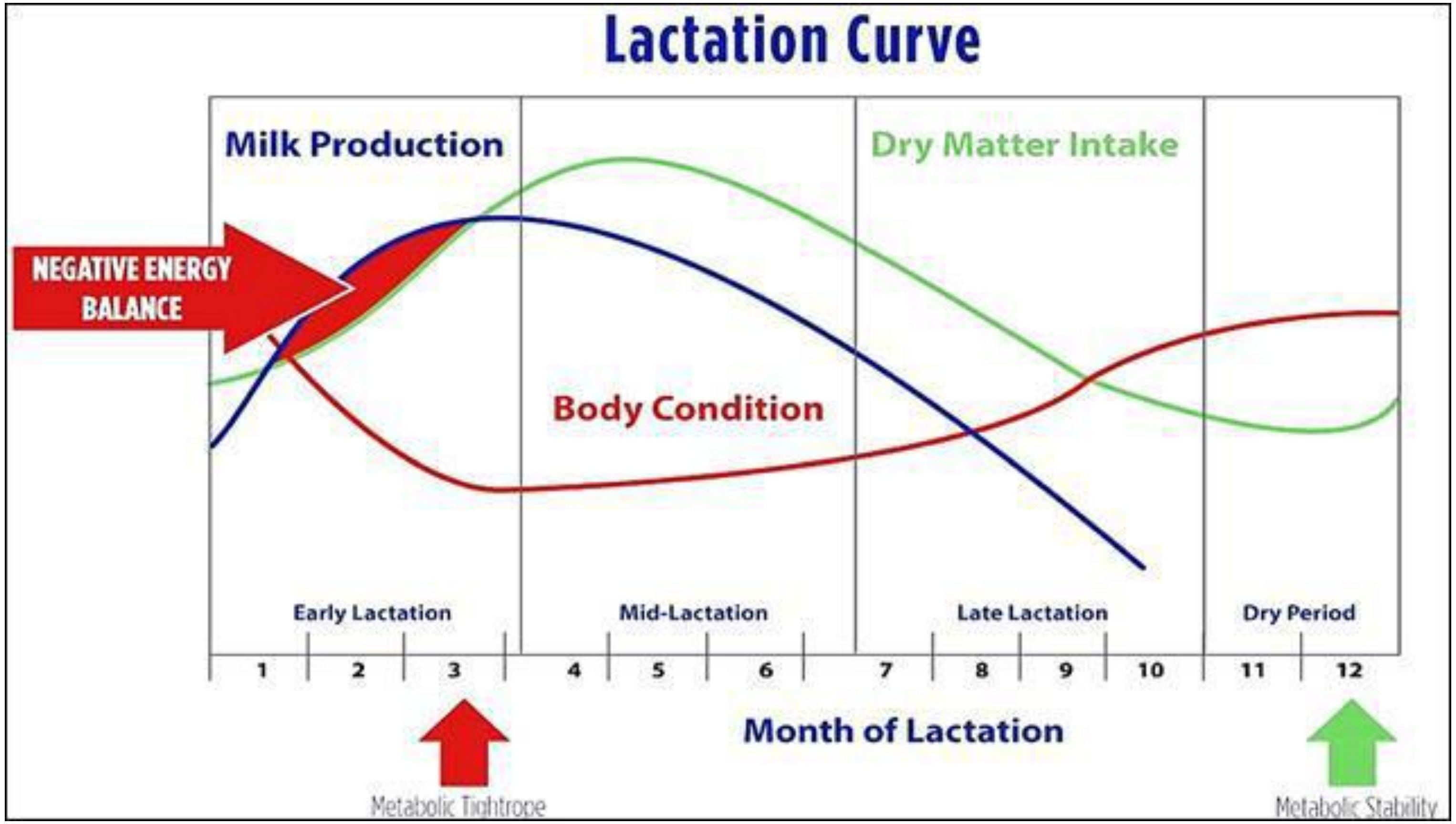
Mid-lactation period
(May sample collection)



Late-lactation period
(October sample collection)

(McHugh et al., 2020)

Samples were collected during two “seasons”



(Tommy, 2020)

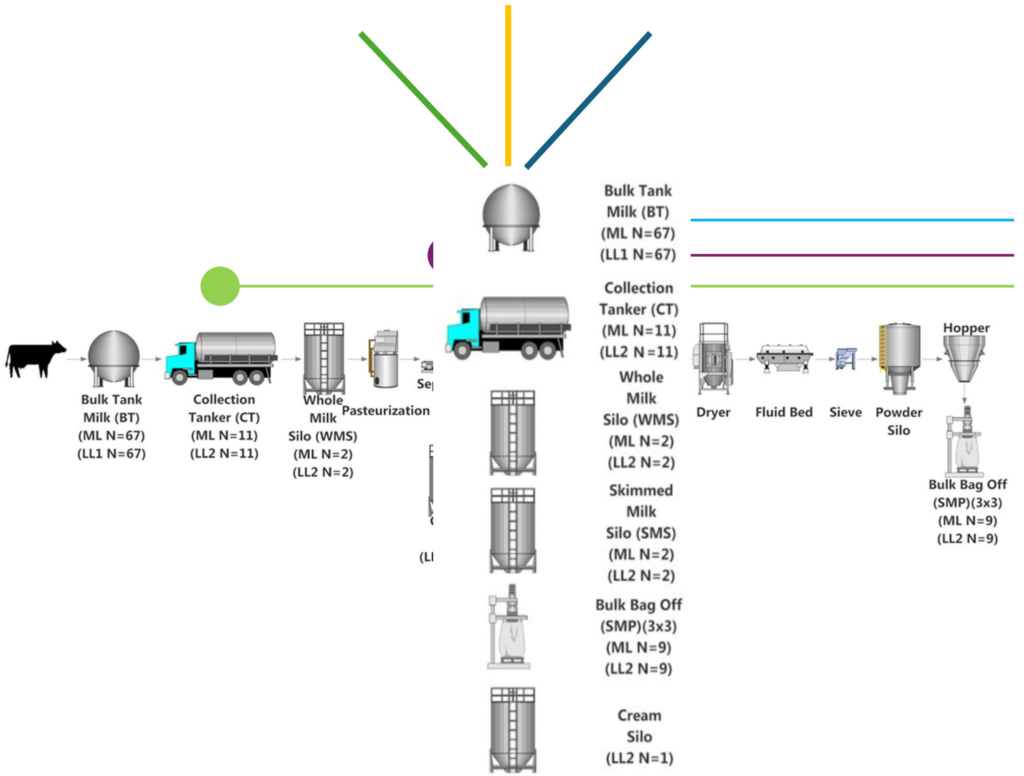
Analysis of microbial communities across dairy manufacturing pipelines help inform industry practices and keep consumers safe!

Despite tightly regulated processing:

- Microorganisms enter and persist in dairy at several stages of the processing chain (Wu et al., 2018)
- Environmental factors can influence the microbiota in the raw milk ahead of processing (Doyle et al., 2018)

The impacts of processing on the milk microbiota have been studied…

- But more needs to be done in order provide a complete picture!



(McHugh et al., 2020)

How does the microbial diversity of dairy samples change throughout the skim milk powder manufacturing process?

- **Hypothesis #1**: Alpha diversity decreases as the raw milk is processed to become skim milk powder.
- **Hypothesis #2**: Sample type is the primary driver shaping the microbial beta diversity in this dataset.
- **Hypothesis #3**: Mid-lactation skim milk powder will be enriched with thermophilic genera.

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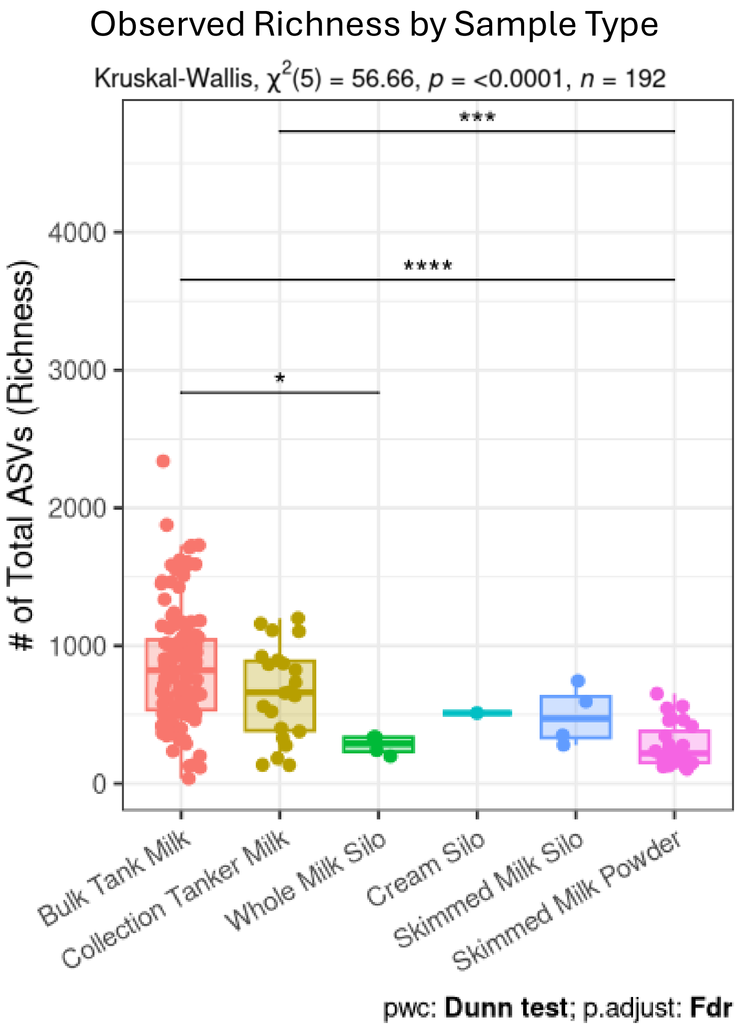
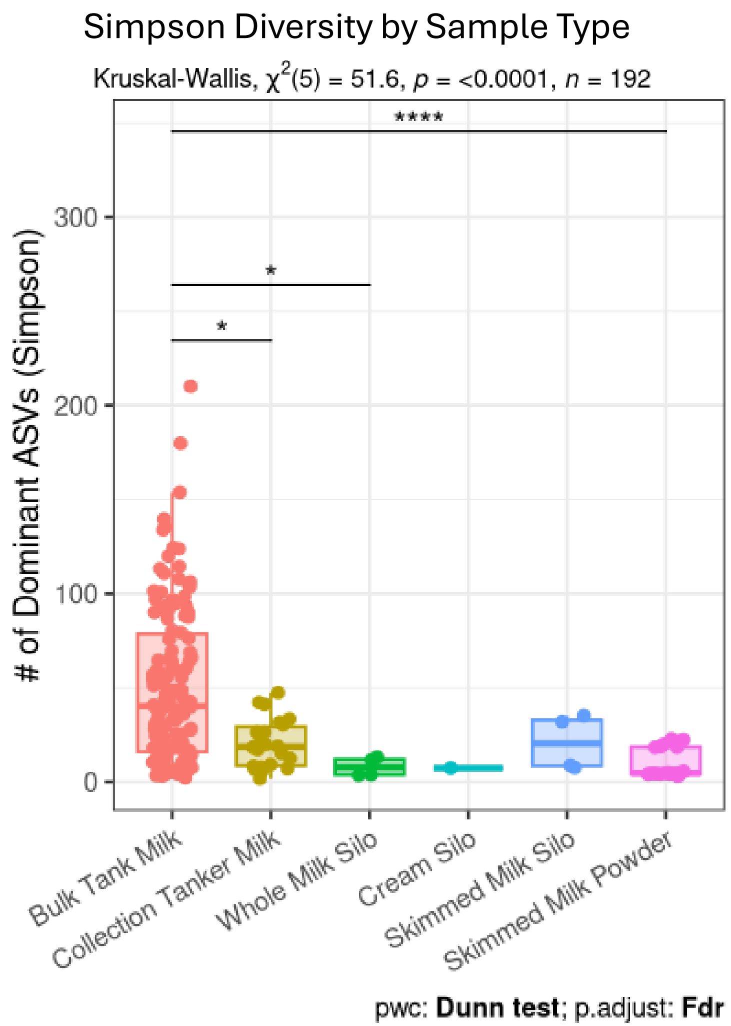
More processed samples, with the exception of skim milk silo samples, have lower measures of richness and Simpson diversity





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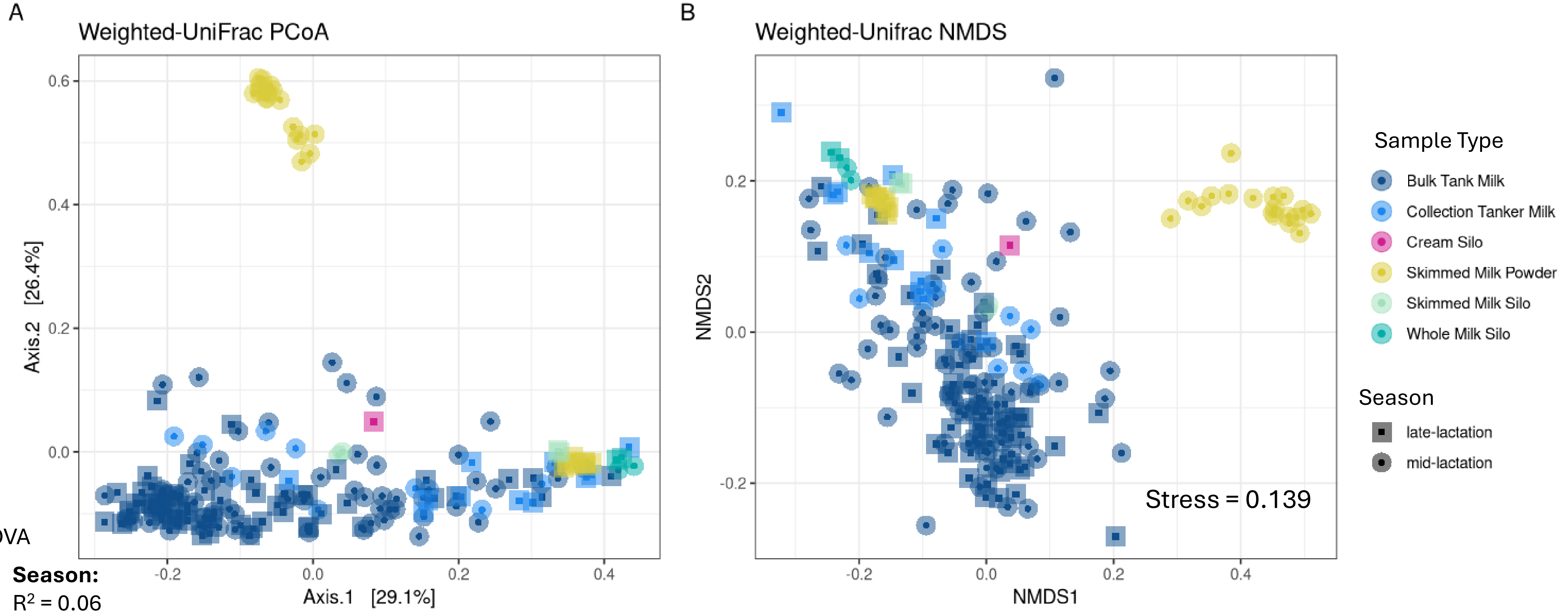
Sample type and season both significantly influence the variation in microbial community composition

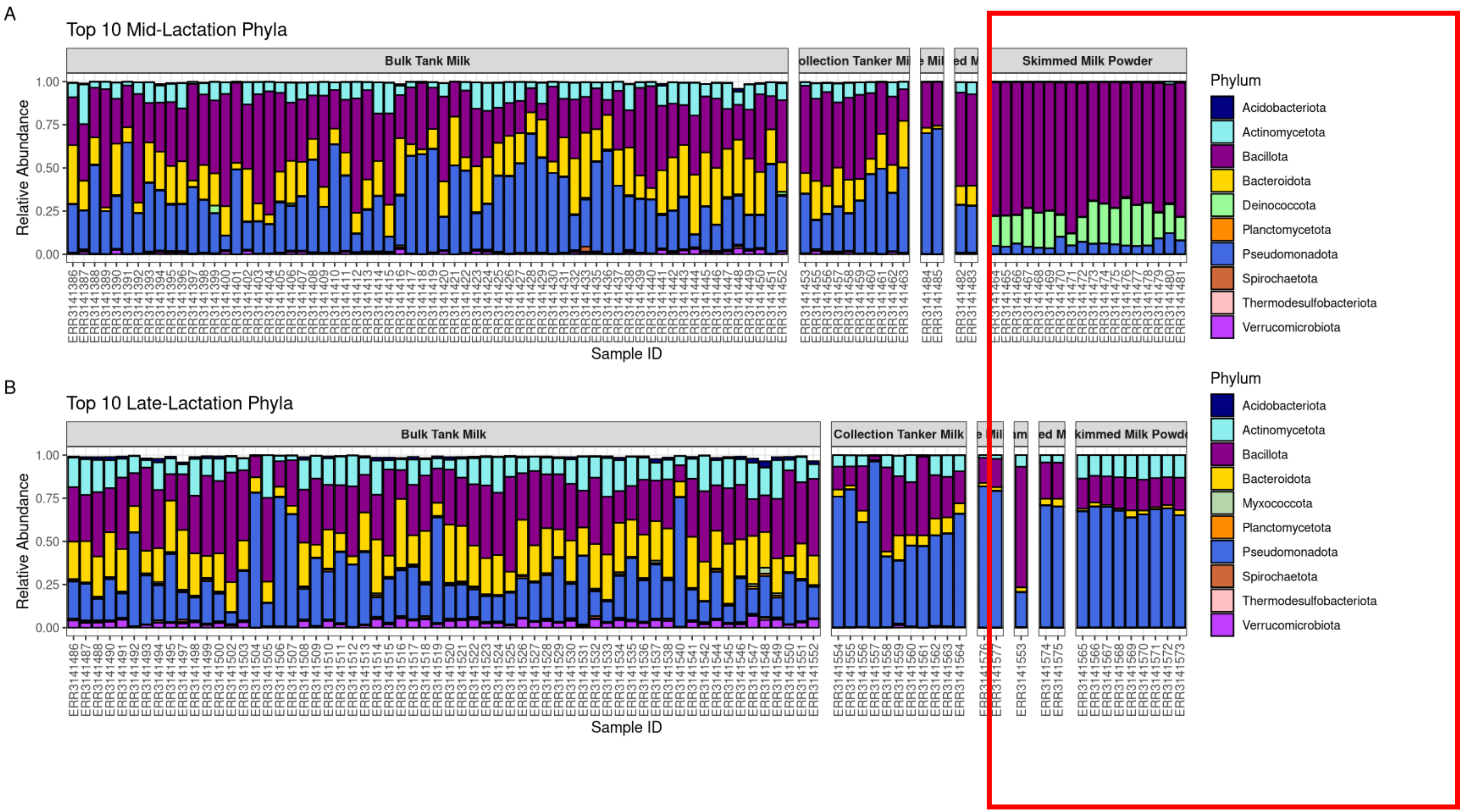


PERMANOVA

Sample Type:
$R^2 = 0.27$
$F = 16.9$
$p < 0.001$

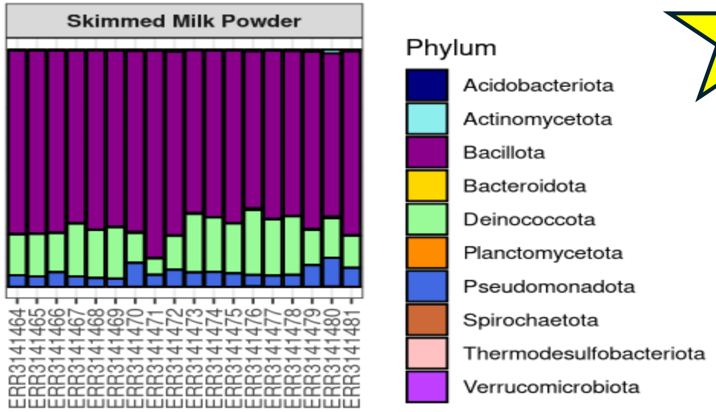
Season:
$R^2 = 0.06$
$F = 17.6$
$p < 0.001$

Microbial composition differs across skim milk powder samples collected in different seasons



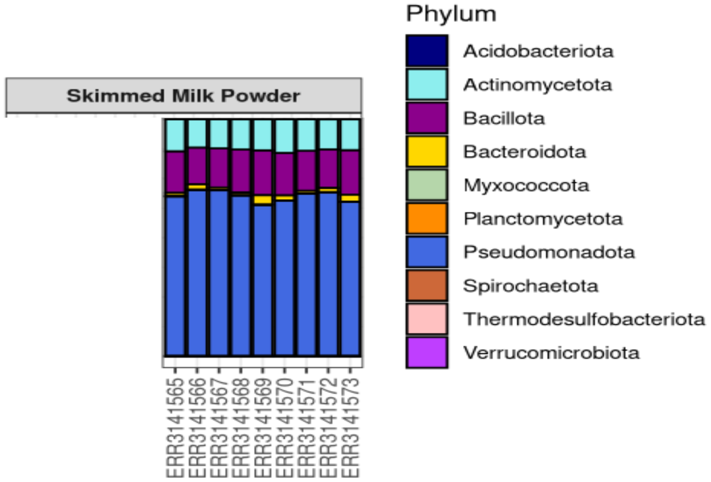
Microbial composition differs across skim milk powder samples collected in different seasons

Mid-lactation
(10 most abundant phyla)



- Sudden increase in relative abundance of Deinococcota

Late-lactation
(10 most abundant phyla)



- Pseudomonadota remains dominant

How does microbial diversity change throughout the skim milk powder manufacturing process?

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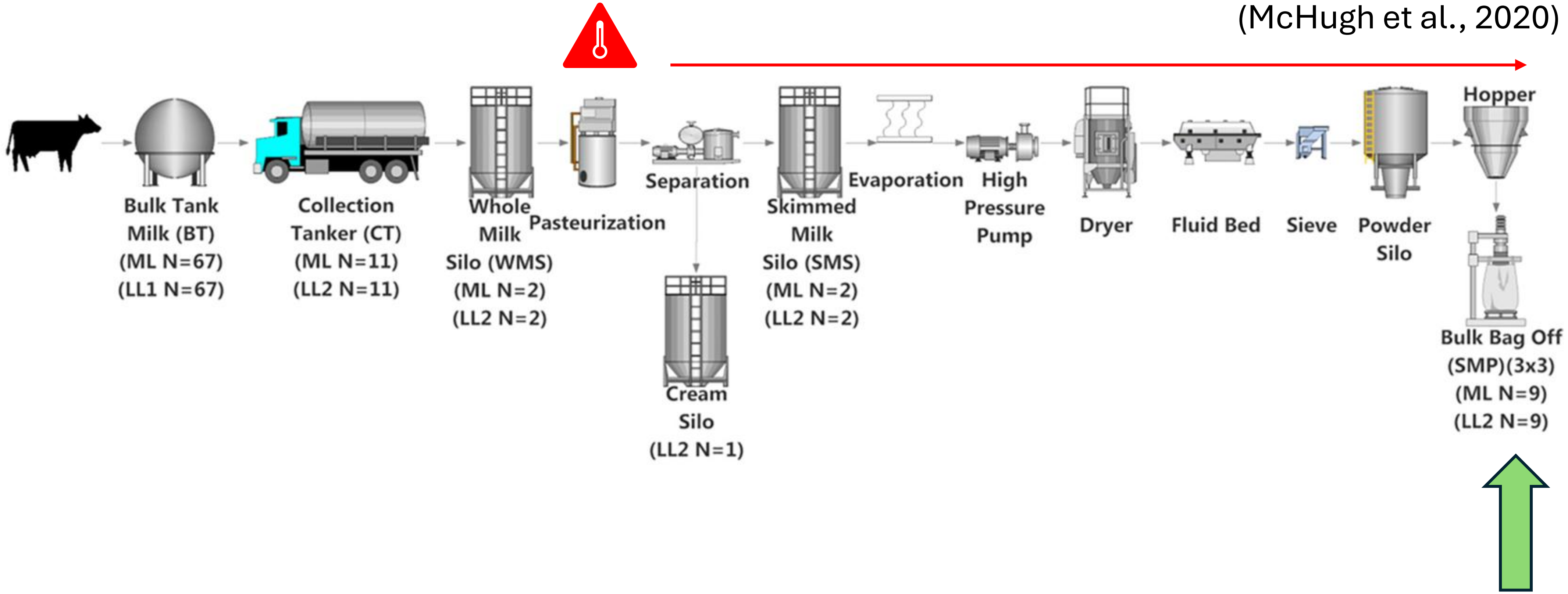
- **Hypothesis #3**: Mid-lactation skim milk powder will be enriched with thermophilic genera.

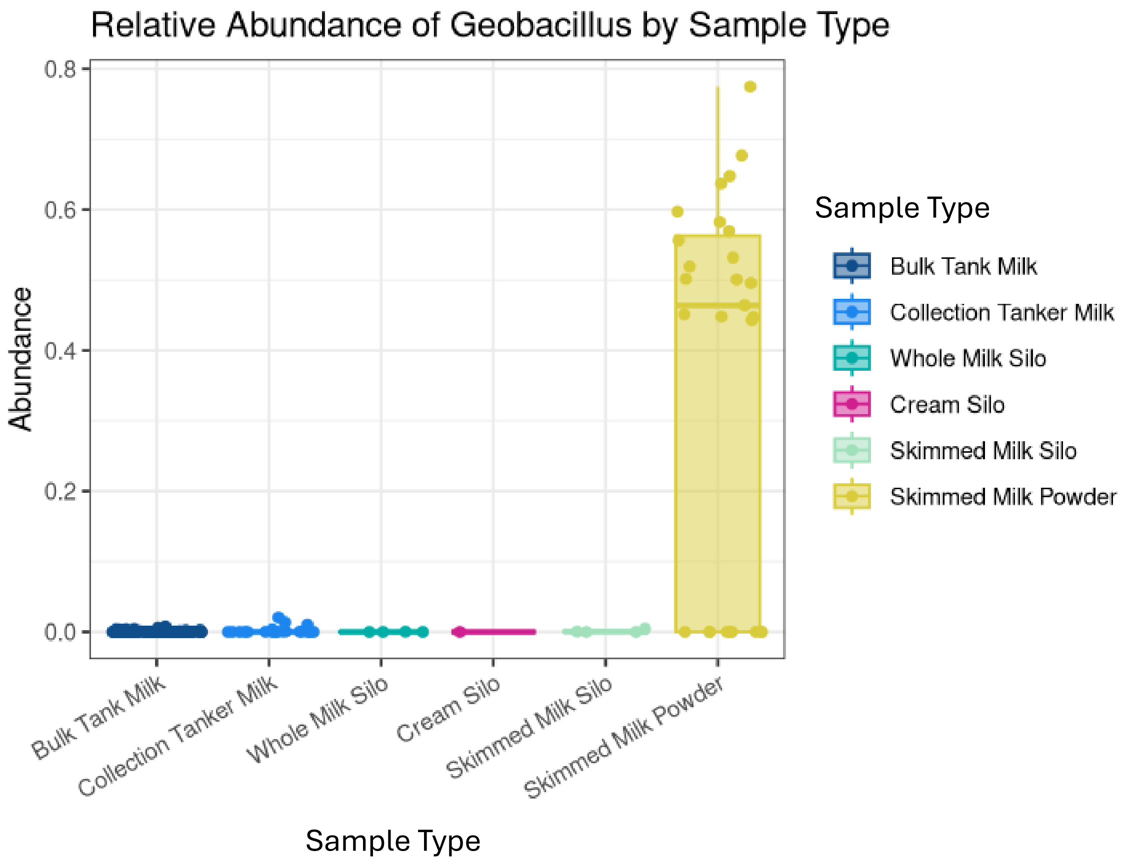
How does microbial diversity change throughout the skim milk manufacturing process?

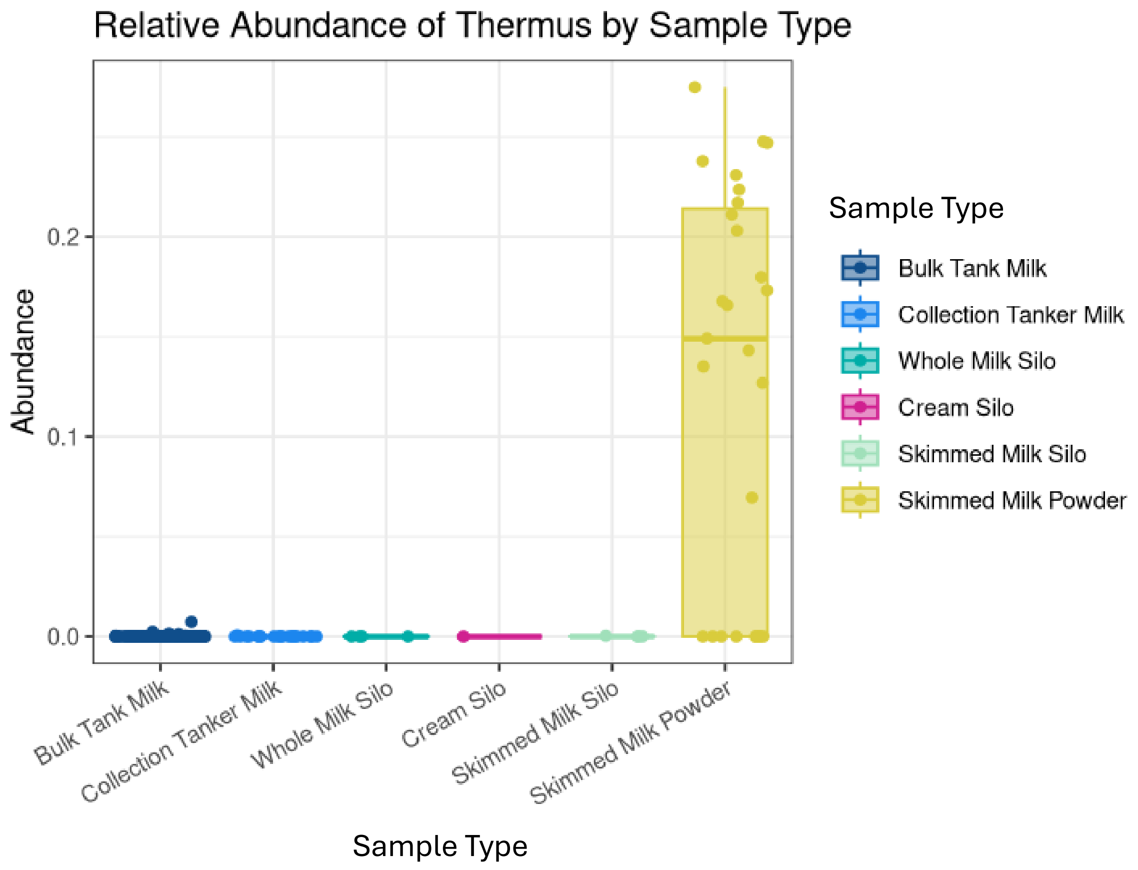
(McHugh et al., 2020)



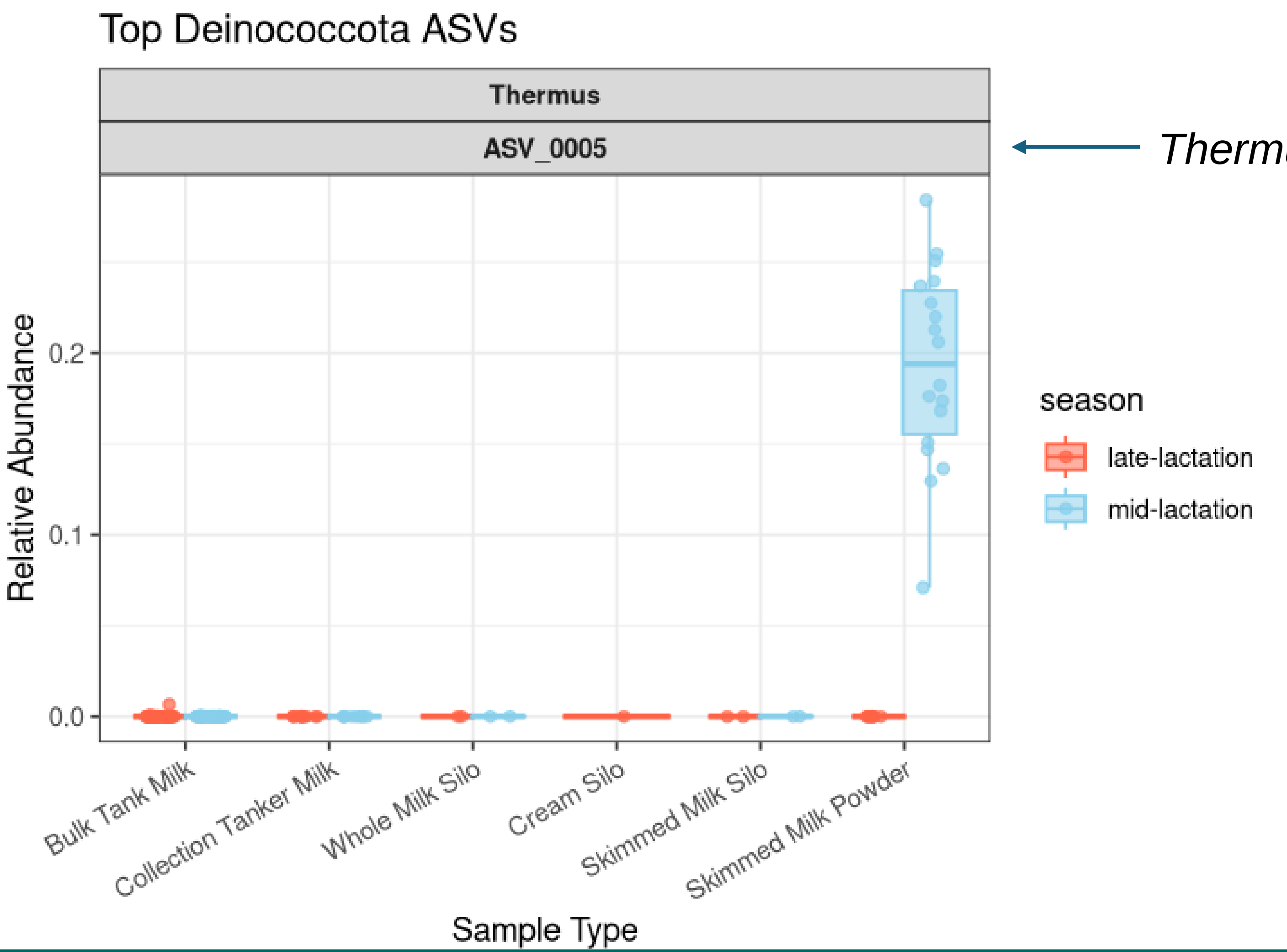
- **Hypothesis #3**: Skim milk powder will be enriched with thermophilic genera.

Skim milk powder had increased relative abundance of both *Thermus* and *Geobacillus*





Thermus was enriched in mid-lactation samples, but not late-lactation samples



Thermus thermophilus

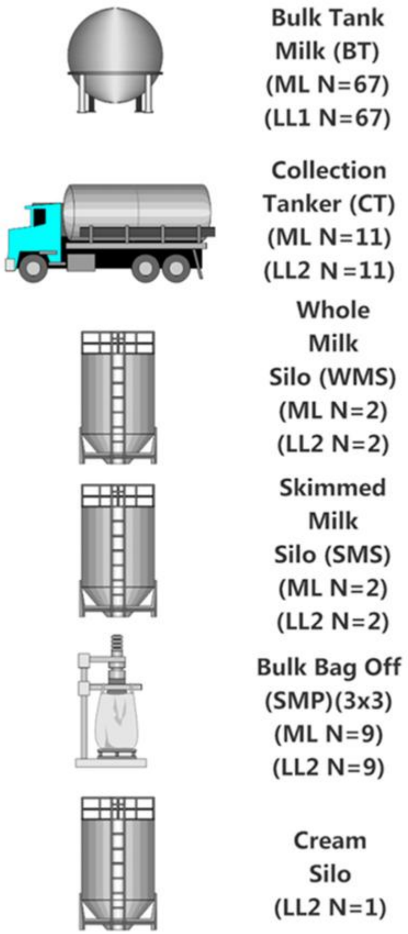
Conclusions

- How does microbial diversity of dairy samples change throughout the skim milk powder manufacturing process?

  - More processed samples have lower measures of richness and Simpson diversity

  - Microbial composition differs across skim milk powder samples collected in different seasons

  - Mid-lactation skim milk powder samples were enriched for the thermophilic genera *Thermus* and *Geobacillus*



(McHugh et al., 2020)

Future Directions

- Further explore:
  - Presence/abundance *Cronobacter*
  - Seasonal differences in microbial communities within sample types

- Utilize 16S rRNA amplicon sequencing in combination with absolute counts to help inform and improve dairy industry processing practices
  - Routine microbiology testing
  - Improved detection of sources of contamination
  - Tracking microbes throughout processing pipelines

Thank you **dairy** much!

References

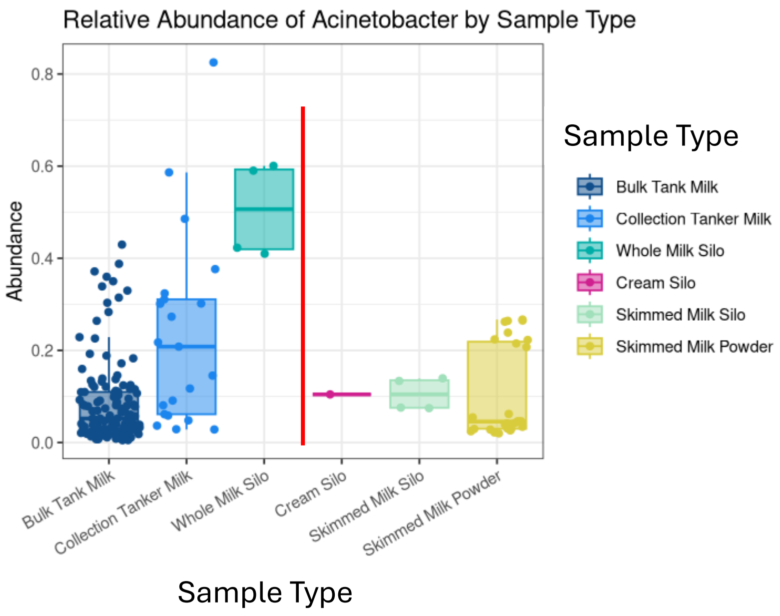
1. Doyle CJ, Gleeson D, O’Toole PW, Cotter PD. 2017. Impacts of seasonal housing and teat preparation on raw milk microbiota: a high throughput sequencing study. Appl Environ Microbiol 83:e02694-16. https://doi.org/10.1128/AEM.02694-16.
2. McHugh, A. J., Feehily, C., Fenelon, M. A., Gleeson, D., Hill, C., & Cotter, P. D. (2020). Tracking the dairy microbiota from farm bulk tank to skimmed milk powder. *mSystems*, *5*(2). https://doi.org/10.1128/msystems.00226-20
3. Tommy. (2020, September 2). *Feeding the high-producing Dairy Cow*. Pacer Technology. https://www.pacer.technology/feeding-the-high-producing-dairy-cow/
4. Wu S, Jiang Y, Lou B, Feng J, Zhou Y, Guo L, Forsythe SJ, Man C. 2018. Microbial community structure and distribution in the air of a powdered infant formula factory based on cultivation and high-throughput se quence methods. J Dairy Sci 101:6915–6926. https://doi.org/10.3168/jds .2017-13968

My scientific question: How does microbial diversity change throughout the skim milk manufacturing process?

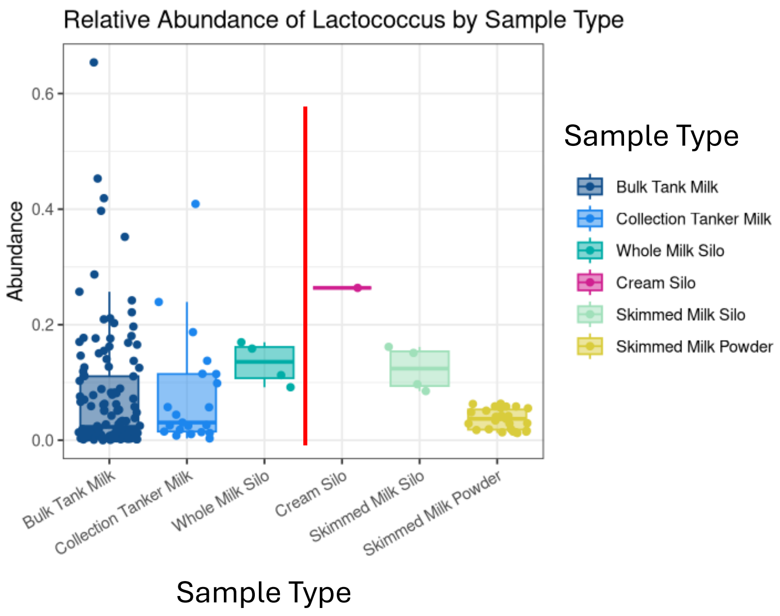
- **Hypothesis #1**: Alpha diversity decreases as the raw milk is processed to become skim milk powder.
- **Hypothesis #2**: Sample type (stage in processing) is the primary driver shaping the microbial beta diversity in this dataset.
- **Hypothesis #3**: Refrigerated storage (pre-pasteurization) will result in an increased relative abundance of psychrotrophic spoilage-associated bacteria. After pasteurization, these psychrotrophic bacteria will diminish.
- **Hypothesis #4**: Skim milk powder will be enriched with thermophilic genera.

Refrigerated storage (pre-pasteurization) results in an increased relative abundance of *Acinetobacter*, *Lactococcus* and *Pseudomonas*

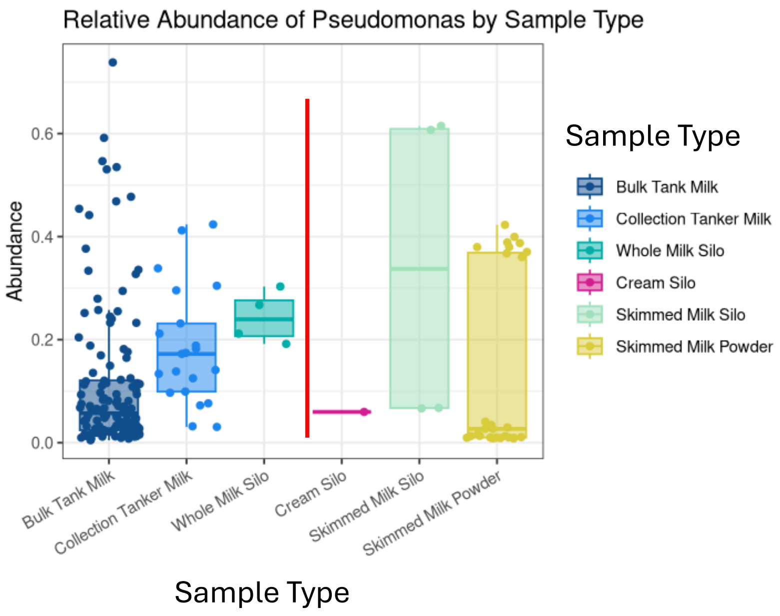
Acinetobacter



Lactococcus

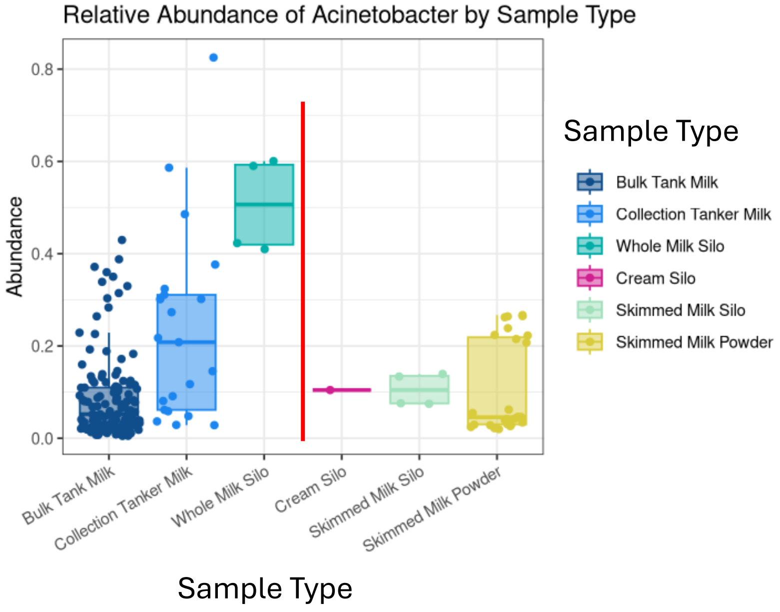


Pseudomonas

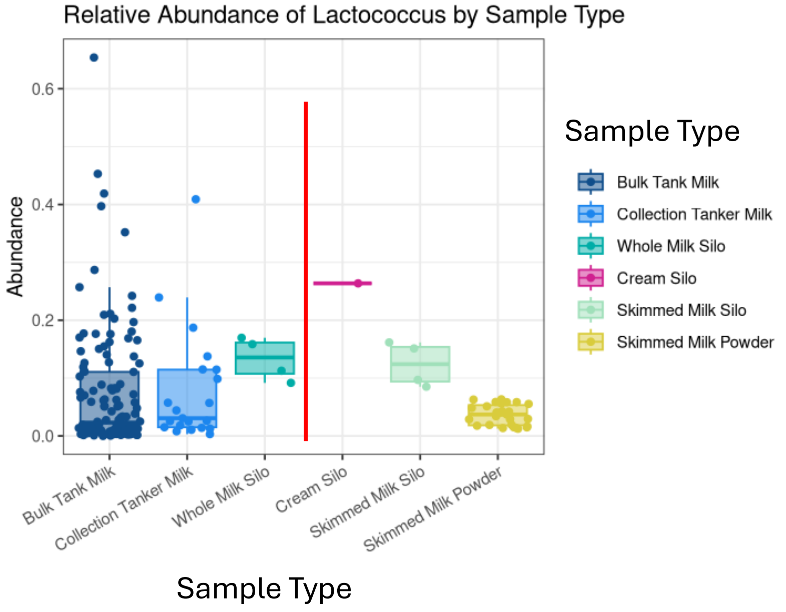


The increased relative abundance of these psychrotrophic, food spoilage-associated genera can persist after pasteurization

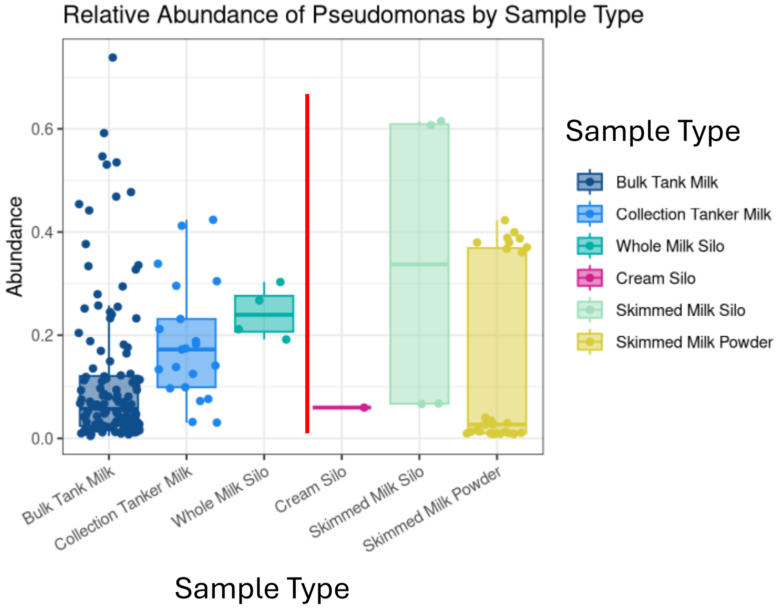
Acinetobacter



Lactococcus



Pseudomonas



This paper connects to my rotation project in the Wiedmann Lab

Cronobacter represents a major concern for the dairy industry